

QUESTION 1

a) Simplify $\frac{x}{2} - \frac{4-2x}{3}$ (2)

b) Solve $x^2 - 3x - 10 > 0$ (2)

c) Find a primitive function of (2)

i) $4x - 2$

ii) $\frac{1}{x^3}$

d) Solve $x^3 - x^2 - x + 1 \geq 0$ (3)

e) Find $\lim_{x \rightarrow 3} \frac{4x^3 - 108}{3 - x}$ (3)

QUESTION 2**Start a New Page**

a) Consider the curve $y = 1 + 3x - x^3$ for $-2 \leq x \leq 3$

- i) Find the co-ordinates of the stationery points and determine their nature (3)
- ii) Find any point(s) of inflexion (2)
- iii) Sketch the curve for $-2 \leq x \leq 3$ (2)
- iv) What is the minimum value of y for $-2 \leq x \leq 3$ (1)
- v) For $-2 \leq x \leq 3$, for what values of x is the curve concave up? (1)

b) Find the values of m for which the roots of the quadratic equation

$$2x^2 + x + 5 - m = 0 \quad \text{are}$$

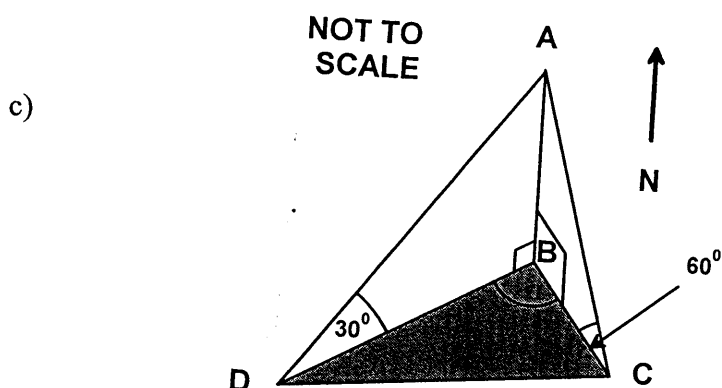
- i) real and equal (2)
- ii) real and reciprocals of each other (1)

QUESTION 3

Start a New Page

- a) Find the co-ordinates of the point which divides the interval AB with $A(-6, 2)$ and $B(4, 7)$ externally in the ratio $3 : 2$ (2)

- b) Solve for x : $\frac{3}{x+5} \leq 1$ (4)



The bearing of C from B is 160° and the bearing of D from B is 220° .
The angle of elevation from D to vertical tower AB is 30° and similarly the angle of elevation of the tower AB from C is 60° . The distance from D to C is 14 metres.

Let $AB = h$ metres

- (i) Explain why $\angle DBC = 60^\circ$. (1)
- (ii) Considering $\triangle ABC$, show that $BC = \frac{h}{\tan 60^\circ}$. (1)
- (iii) Find a similar expression for BD . (1)
- (iv) By using the cosine rule in $\triangle BCD$, show that the height of tower AB is $2\sqrt{21}$ metres exactly. (3)

QUESTION 4**Start a new page**

- a) The gradient function of a curve is $2x - 3$ and the curve passes through $(1, 5)$. Find the equation of the curve. (2)

- b) Find the equation of the parabola with focus $(2, -5)$ and directrix $y = 3$ (3)

Expend?

- c) Given the function $y = \frac{2}{1+x^2}$

- (i) Show that $y = \frac{2}{1+x^2}$ is an even function and describe the graphical significance of an even function. (2)

- (ii) Find $\lim_{x \rightarrow \infty} \frac{2}{1+x^2}$ (1)

- (iii) Calculate the turning point of $y = \frac{2}{1+x^2}$ and determine its nature. (3)

- (iv) Sketch the curve $y = \frac{2}{1+x^2}$ (1)

QUESTION 5**Start a New Page**

a) Find the domain of $y = \frac{1}{\sqrt{(9 - x^2)}}$ (2)

b) The volume, V , of water in a dam at time t was monitored over a period of time. When monitoring began, the dam was 75% full. During the monitoring period, the volume decreased at an increasing rate due to a long period of drought.

i) What does this tell us about V' and V'' ? (2)

ii) Sketch a graph of V against t . (1)

c) Given the series

$2 + 4 + 6 + \dots + 1000$, find

(i) the 21st term (1)

(ii) the sum of the series (2)

d) Use Mathematical Induction to prove that, for all positive integers n :

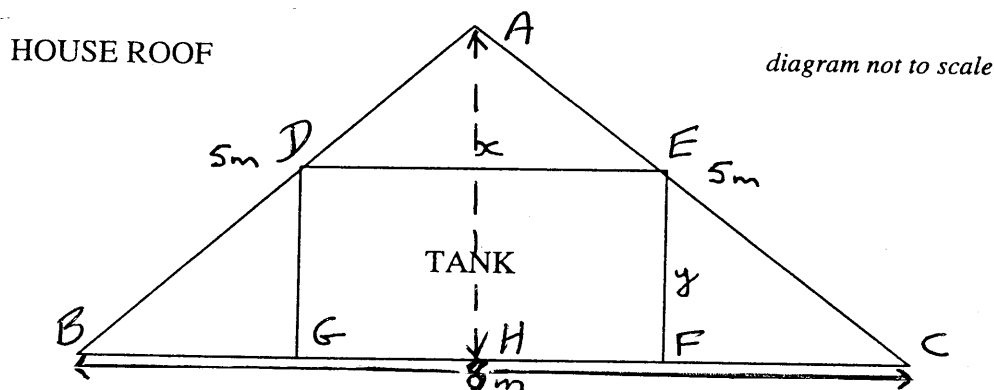
$$1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1) \quad (4)$$

QUESTION 6

a) Solve $(2^x)^2 + 7(2^x) - 8 = 0$ (2)

b) Evaluate $\sum_{n=4}^{50} (3n + 2)$ (2)

c) The diagram below show a rectangular water tank, x metres wide, y metres high and 1.5 metres long which fits exactly into the roof of a house. A cross section of the roof is in the shape of an isosceles triangle with base 8 metres and equal sides 5 metres in length.



- Copy the diagram onto your answer sheet.
- Explain why the roof of the house is 3 metres high. (1)
- Show that $FC = 4 - \frac{x}{2}$ (1)
- Using the fact that $\triangle EFC \parallel \triangle AHC$, show that $y = \frac{1}{8}(24 - 3x)$ (2)
- Show that the volume of the tank is given by:

$$V = \frac{3}{16} x (24 - 3x) \quad (1)$$

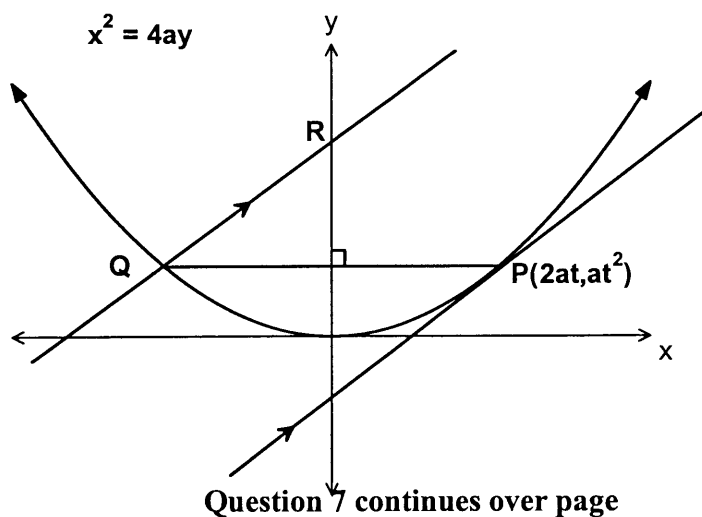
- Calculate the maximum volume of the tank. (3)

QUESTION 7

(a) A stone dropped from the top of a 245m high cliff falls 5 metres in the first second, 15 metres in the second second, 25 metres in the third and so on.

- (i) Find a formula for the distance fallen after n seconds (2)
- (ii) How long does it take before the stone hits the ground? (1)

(b) $P(2at, at^2)$ lies on $x^2 = 4ay$. PQ is perpendicular to the axis, and the line through Q parallel to the tangent at P meets the y axis in R .



- (i) Show that the equation of the tangent at P is $tx - y - at^2 = 0$. (2)
- (ii) Write down the coordinates of Q in terms of t . (1)
- (iii) Find the equation of QR . (2)
- (iv) Find the coordinates of R . (1)
- (v) Hence find the cartesian equation of the locus of midpoints of QR (3)

End of paper!

Suggested Solutions

Comments

Question (1)

$$a) \frac{x}{2} - \frac{4-2x}{3} = \frac{3x - 2(4-2x)}{6} \quad (1)$$

$$= \frac{3x - 8 + 4x}{6} \quad (1)$$

$$= \frac{7x - 8}{6}$$

$$b) x^2 - 3x - 10 > 0 \quad (1)$$

$$(x-5)(x+2) > 0 \quad (1)$$

$$x < -2, x > 5$$

$$c) i) 4\frac{x^2}{2} - 2x + c = 2x^2 - 2x + c \quad |$$

$$ii) \frac{x}{-2} + c = \frac{1}{-2x^2} + c \quad |$$

$$d) x^3 - x^2 - x + 1 \geq 0$$

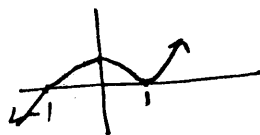
$$x^2(x-1) - 1(x-1) \geq 0 \quad (1)$$

$$(x-1)(x^2-1) \geq 0 \quad (1)$$

$$(x-1)(x-1)(x+1) \geq 0$$

$$(x-1)^2(x+1) \geq 0 \quad (1)$$

$$x \geq -1$$



$$e) \lim_{x \rightarrow 3} \frac{4x^3 - 108}{3-x} = \lim_{x \rightarrow 3} \frac{4(x^3 - 27)}{3-x} \quad (1)$$

$$= \lim_{x \rightarrow 3} \frac{4(x-3)(x^2+3x+9)}{3-x} \quad (1)$$

$$= \lim_{x \rightarrow 3} \frac{-4(3-x)(x^2+3x+9)}{3-x}$$

$$= -4 \times 27 \quad (1)$$

$$= -108$$

Suggested Solutions

Comments

Question (2)

a) $y = 1 + 3x - x^3$

i) $y' = 3 - 3x^2$

stationary points $y' = 0$

$$0 = 3(1 - x^2)$$

$$= 3(1 - x)(1 + x)$$

$$x = 1, -1$$

at $x = 1$, $y = 1 + 3(1) - (1)^3$
 $= 3$

at $x = -1$, $y = 1 + 3(-1) - (-1)^3$
 $= -1$

Stationary points are $(1, 3)$ and $(-1, -1)$

$$y'' = -6x$$

at $x = 1$, $y'' = -6 < 0 \therefore$ Maximum turning point at $(1, 3)$

at $x = -1$, $y'' = 6 > 0 \therefore$ Minimum turning point at $(-1, -1)$

ii) Points of inflexion: $y'' = 0$

$$-6x = 0$$

$$x = 0$$

x	-1	0	1
y''	6	0	-6

change in concavity.

Point of inflexion at $(0, 1)$

iii) At $x = -2$, $y = 1 + 3(-2) - (-2)^3$
 $= 3 \quad (-2, 3)$

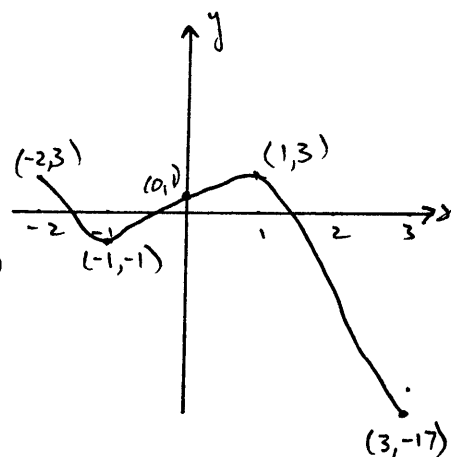
at $x = 3$, $y = 1 + 3(3) - (3)^3$
 $= -17 \quad (3, -17)$

iv) Minimum value is -17

(1)

v) $-2 \leq x < 0$

(1)



b) i) $\Delta = (1)^2 - 4 \times 2 \times (5 - m) = 0$
 $1 - 40 + 8m = 0$

$$8m = 39$$

$$m = \frac{39}{8}$$

(2)

ii) $\alpha\beta = 1$

$$\frac{c}{a} = 1$$

$$\frac{5-m}{2} = 1$$

$$5 - m = 2$$

$$m = 3$$

(3)

1 for finding points only

1 for $(0, 1)$

1 for testing concavity

1 for $(-2, 3)$ and $(3, -17)$

1 for $\Delta = 0$

Question (3)

$$a) \left(\frac{3 \times 4 - 2 \times -6}{1}, \frac{3 \times 7 - 2 \times 2}{1} \right) = (24, 17)$$

$$b) \frac{3}{x+5} (x+5)^2 \leq 1(x+5)^2 \quad (1) \quad x \neq -5$$

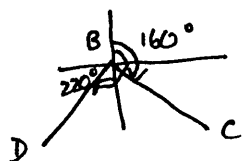
$$0 \leq (x+5)^2 - 3(x+5) \quad (1)$$

$$(x+5)[(x+5) - 3] \geq 0$$

$$(x+5)(x+2) \geq 0 \quad (1) \quad \begin{array}{c} \leftarrow \quad \rightarrow \\ -5 \quad -2 \end{array}$$

$$x \leq -5, x \geq -2 \quad (1)$$

c) i)



$$\therefore \angle DBC = 220^\circ - 160^\circ = 60^\circ \quad (1)$$

ii)



$$\tan 60^\circ = \frac{h}{BC}$$

$$BC = \frac{h}{\tan 60^\circ}$$

(1)

$$iii) BD = \frac{h}{\tan 30^\circ} \quad (1)$$

$$iv) 14^2 = \left(\frac{h}{\tan 60^\circ} \right)^2 + \left(\frac{h}{\tan 30^\circ} \right)^2 - 2 \left(\frac{h}{\tan 60^\circ} \right) \left(\frac{h}{\tan 30^\circ} \right) \cos 60^\circ \quad (1)$$

$$196 = \left(\frac{h}{\sqrt{3}} \right)^2 + \left(\frac{h}{\frac{1}{\sqrt{3}}} \right)^2 - 2 \left(\frac{h}{\sqrt{3}} \right) (\sqrt{3}h) \cdot \frac{1}{2}$$

$$= \frac{h^2}{3} + 3h^2 - h^2 \quad (1)$$

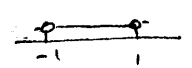
$$196 = \frac{7h^2}{3}$$

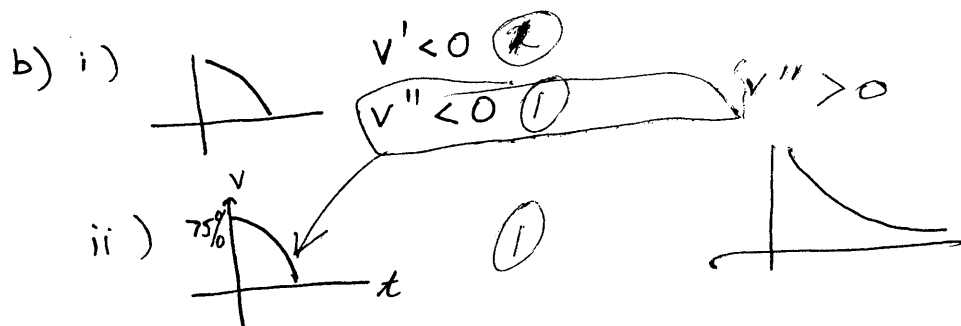
$$84 = h^2$$

$$h = \sqrt{21 \times 4}$$

$$= 2\sqrt{21} \text{ metres} \quad (1)$$

Question (5)

a) $y = \frac{1}{\sqrt{9-x^2}}$
 $= \frac{1}{\sqrt{(3-x)(3+x)}}$ $x \neq 3, -3$ ①
 $(3-x)(3+x) > 0$  ①
 $-3 < x < 3$



c) i) $a=2, d=2. T_n = a + (n-1)d$ ①
 $T_{21} = 2 + 20 \times 2$
 $= 42$

ii) $1000 = 2 + (n-1)2$
 $= 2n$ ①
 $n = 500$
 $S_n = \frac{n}{2}(a+l)$
 $S_{500} = \frac{500}{2}(2+1000)$ ①
 $= 250500$

d) Step 1: To prove true for $n=1$
 $LHS = 1^2 = 1$ $RHS = \frac{1}{6} \times 1(1+1)(2 \times 1+1)$ ①
 $= 1$ $= \frac{1}{6} \times 2 \times 3$
 $= 1$
 $= LHS$

$\therefore$ True for $n=1$

Step 2: Assume true for $n=k$ ①
 i.e. $1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{1}{6} k(k+1)(2k+1)$

To prove true for $n=k+1$
 i.e. $1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{1}{6} (k+1)(k+2)(2k+3)$

$LHS = \frac{1}{6} k(k+1)(2k+1) + (k+1)^2$
 $= \frac{1}{6} (k+1) [k(2k+1) + 6(k+1)]$
 $= \frac{1}{6} (k+1) [2k^2 + k + 6k + 6]$
 $= \frac{1}{6} (k+1) [2k^2 + 7k + 6]$
 $= \frac{1}{6} (k+1) [2k^2 + 4k + 3k + 6]$
 $= \frac{1}{6} (k+1) [2k(k+2) + 3(k+2)]$
 $= \frac{1}{6} (k+1)(k+2)(2k+3)$
 $= RHS$

Step 3

Suggested Solutions

Comments

Question (4)

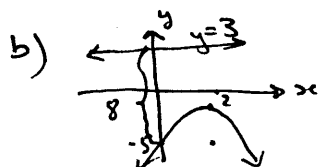
a) $y' = 2x - 3$

$y = x^2 - 3x + c$ ①

$x=1, y=5, 5 = (1)^2 - 3(1) + c$

$7 = c$ ①

$\therefore y = x^2 - 3x + 7$

Vertex $(2, -1)$

$a = 4$

$(x-h)^2 = -4a(y-k)$

$(x-2)^2 = -16(y+1)$

1 for vertex

1 for a

1 for equation/format

c) i) $f(x) = \frac{2}{1+x^2}$

$f(-x) = \frac{2}{1+(-x)^2}$

$= \frac{2}{1+x^2}$ ①

 $= f(x)$. $\therefore$ even function.

Symmetrical in the y-axis ①

ii) $\lim_{x \rightarrow \infty} \frac{2}{1+x^2} = \lim_{x \rightarrow \infty} \frac{\frac{2}{x^2}}{\frac{1}{x^2} + \frac{x^2}{x^2}}$

$= \frac{0}{0+1}$ ①

$= 0$

iii) $y = \frac{2}{1+x^2}$

$y' = \frac{(1+x^2) \cdot 0 - 2 \cdot (2x)}{(1+x^2)^2}$

$= \frac{-4x}{(1+x^2)^2}$

stationary points $y' = 0$

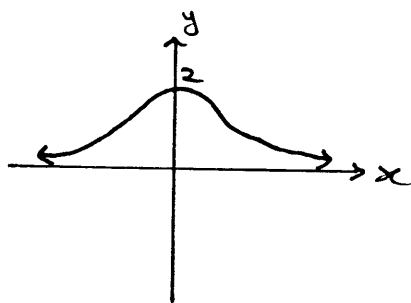
$0 = \frac{-4x}{(1+x^2)^2}$

$x = 0$. $(0, 2)$

x	-1	0	1
y	1	0	-

 $\therefore$ Maximum turning point at $(0, 2)$

iv)



①

1 - quotient rule/product rule

1 $(0, 2)$

1 natural

$y'' = \frac{-4(1+x^2)^2 - 4x[2(1+x^2) \cdot 2x]}{(1+x^2)^4}$

$= \frac{-4(1+x^2)^2 + 16x^2(1+x^2)}{(1+x^2)^4}$

Question (6)

$$a) (2^x)^2 + 7(2^x) - 8 = 0$$

$$\text{Let } u = 2^x$$

$$u^2 + 7u - 8 = 0$$

$$(u+8)(u-1) = 0$$

$$u = -8, 1$$

$$2^x = -8, 1$$

$$x = 0 \quad (\text{no real solution for } 2^x = -8)$$

(2)

1 for $u = -8, 1$

1 for answer

$$b) \sum_{n=4}^{50} (3n+2) = 14 + 17 + 20 + \dots + 152$$

$$S_n = \frac{n}{2}(a+l)$$

$$S_{47} = \frac{47}{2}(14+152)$$

$$= 3901$$

(2)

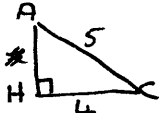
$$n = 47$$

$$a = 14$$

$$d = 3$$

$$l = 152$$

$14+17+\dots+152 = 3901$ marks.
Needed to show some idea
of sum of arithmetic
series.

c) ii)  By Pythagoras' Rule,

$$5^2 = AH^2 + 4^2$$

$$AH = 3.$$

(1)

iii) $FC = HC - HF$

$$= 4 - \frac{1}{2}x$$

(1)

iv) $\frac{3}{y} = \frac{4}{4-x/2}$

$$4y = 3(4 - \frac{x}{2})$$

$$4y = 12 - \frac{3x}{2}$$

$$8y = 24 - 3x$$

$$y = \frac{1}{8}(24 - 3x)$$

(2)

v) $V = x \cdot y \cdot 1.5$

$$= 1.5x \cdot \frac{1}{8}(24 - 3x)$$

$$= \frac{3}{16}x(24 - 3x)$$

(1)

vi) $V' = \frac{3}{16}(24 - 3x) + \frac{3}{16}x \cdot -3$

$$= 4\frac{1}{2} - \frac{9}{16}x - \frac{9}{16}x$$

$$= 4\frac{1}{2} - \frac{9}{8}x$$

Let $V' = 0$ for stationary points

$$0 = 4\frac{1}{2} - \frac{9}{8}x$$

$$x = 4$$

$$V'' = -\frac{9}{8} < 0 \therefore \text{Maximum.}$$

$$\therefore \text{Maximum volume} = \frac{3}{16}x \cdot 4 \cdot (24 - 3x) = 9 \text{ m}^3$$

1 for $x = 4$

1 for showing a max

1 for 9 m^3

Suggested Solutions

Comments

Question (7)

a) $5 + 15 + 25 + \dots$

$$\begin{aligned}
 \text{i)} \quad S_n &= \frac{n}{2} \{2a + (n-1)d\} \\
 &= \frac{n}{2} \{10 + (n-1) \cdot 10\} \\
 &= \frac{n}{2} \{10 + 10n - 10\} \\
 &= 5n^2
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad 245 &= 5n^2 \\
 n^2 &= 49 \\
 n &= 7 \quad \therefore \text{after 7 seconds}
 \end{aligned}$$

1 for a and d correct.

if solve (i) = 245 correctly, 1 mark.

①

b) i) $x^2 = 4ay$

$y = \frac{x^2}{4a}$

$y' = \frac{2x}{4a}$

$= \frac{x}{2a}$

at $x = 2at$, $y' = \frac{2at}{2a} = t$

$y - y_1 = m(x - x_1)$

$y - at^2 = t(x - 2at)$

$y - at^2 = tx - 2at^2$

$tx - y - at^2 = 0$

ii) $Q = (-2at, at^2)$

iii) $m = t$ since lines parallel

$y - at^2 = t(x + 2at)$

$y - at^2 = tx + 2at^2$

$tx - y + 3at^2 = 0$

iv) let $x = 0$, $-y + 3at^2 = 0$
 $y = 3at^2$

$R(0, 3at^2)$

v) midpoint of QR = $\left(\frac{-2at+0}{2}, \frac{at^2+3at^2}{2}\right)$
 $= (-at, 2at^2)$

$x = -at$
 $t = -\frac{x}{a}$

$$\begin{aligned}
 y &= 2at^2 \\
 &= 2a\left(-\frac{x}{a}\right)^2 \\
 &= 2a \cdot \frac{x^2}{a^2} \\
 &= \frac{2x^2}{a}
 \end{aligned}$$

$y = \frac{2x^2}{a}$

②

①

②

①

1 for gradient = t

1 if let $x=0$ in (iii) & find y correctly.

1 for midpoint